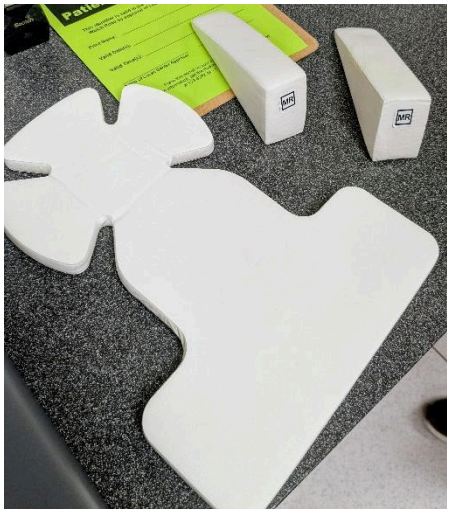
SOP for Impulse Traveling Head Study

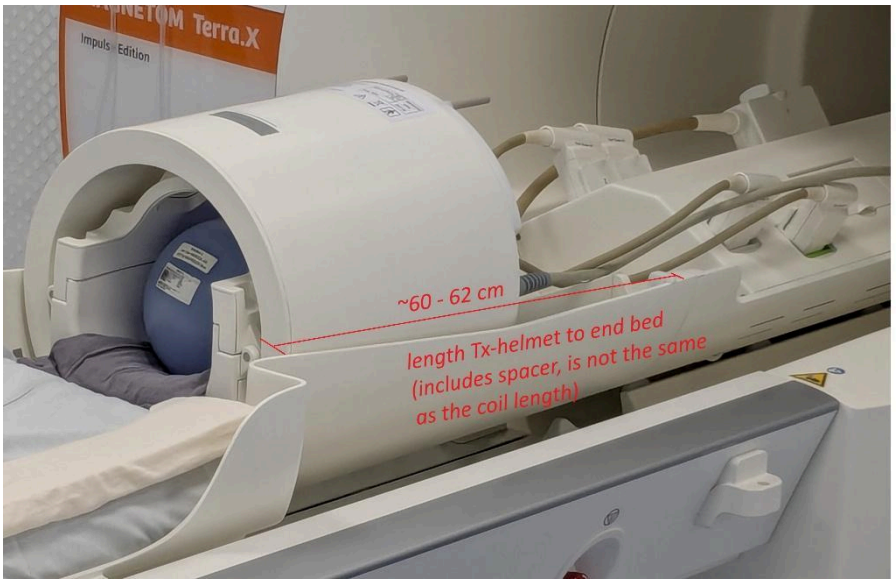
General comments:

- Feedback please at: hendrik.mattern@ovgu.de & saskia.bollmann@uq.edu.au
- includes **fMRI with task** (psychopy)
- acquired **without physio data**
- use the following pillows for the head (see image below):
    - Star-shaped pillow to lie on
    - Triangle-shaped pillows over the ears



Document setting:

- Volunteer name: _____
- Volunteer weight: ____ kg
- Volunteer height: ____ cm
- Operator name: _____
- Scan date: _____
- Distance end patient bed to end transmit coil: _____ cm
  (see image below, 61.5 cm in Stanford, ideally measured by Hendrik)



Notes on PsychoPy:

- There is a dedicated docx explaining how to open and run the tasks

Notes on FOV planning:

- Please obey the following rules and adhere to them at all times when planning the slice/FOV positioning (shifts & angulation):
  - Not allowed when labeled as NO PLANNING
  - Only allowed when the AutoAlign obviously failed; labeled as DOUBLECHECKING
  - Has to be done manually when labeled as MANUAL PLANNING
  - Done via COPYREFERENCE

Scanning:

1. **Localizer** `[strict SAG & in ISO]` NO PLANNING

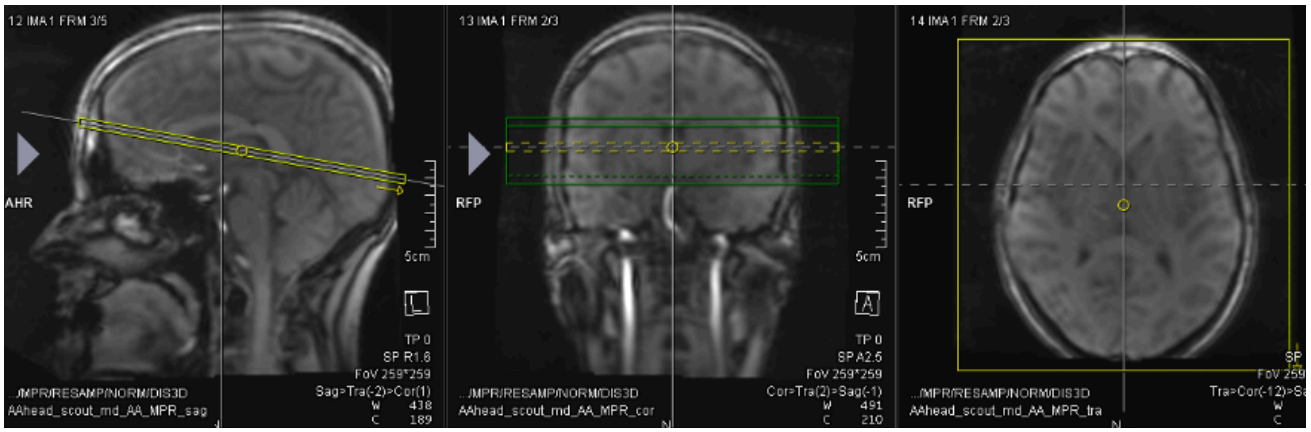
2. **AAhead_scout_md** `[strict SAG & in ISO]` NO PLANNING
   ☐ FYI: used for autoalign, back of the brain might be missing, use localizer if in doubt when planning manually

3. **Manual adjustments for B0 map etc.**
   - FWHM before shimming: _____ Hz
   - Run shim (2x GRE Brain with measure, calculate, apply)
   - FWHM after shimming: _____ Hz

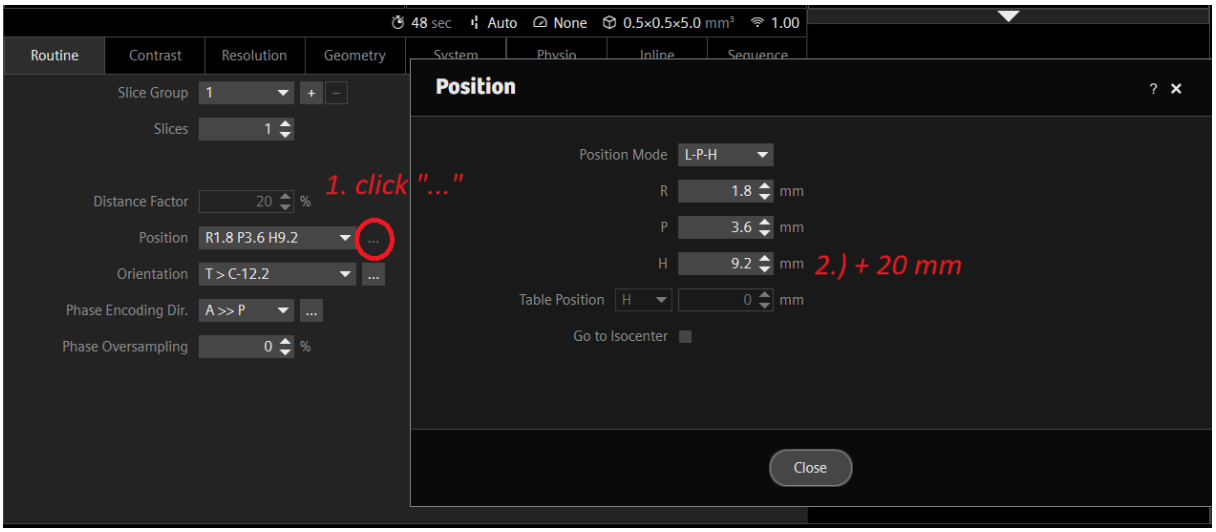
4. **gre_b0map_4iso_sag** `[strict SAG & in ISO]` NO PLANNING

5. **dzne_b1map_5iso_sag** `[strict SAG & in ISO]` NO PLANNING
   ☐ copy adjustment volume from gre_b0map_4iso_sag

6. **gre_FA10_acpc** `[ACPC via AutoAlign; screenshot only for DOUBLECHECKING]`
   - Reference voltage: _____ V
     ☐ save screenshot



7. **gre_FA0_acpc** `[ACPC via COPYREFERENCE]`
   ☐ doublecheck RefVoltage = 0V
   ☐ confirm RefVoltage when asked



gre_FA10_up [corpus callosum, MANUAL PLANNING]
- ☐ use ACPC angulation by copying slice position from gre_FA10_acpc
- ☐ click "…" behind position and add 20 mm to the "H" field (FYI: if "F" subtract 20 mm)
- ☐ save screenshot of FOV planning

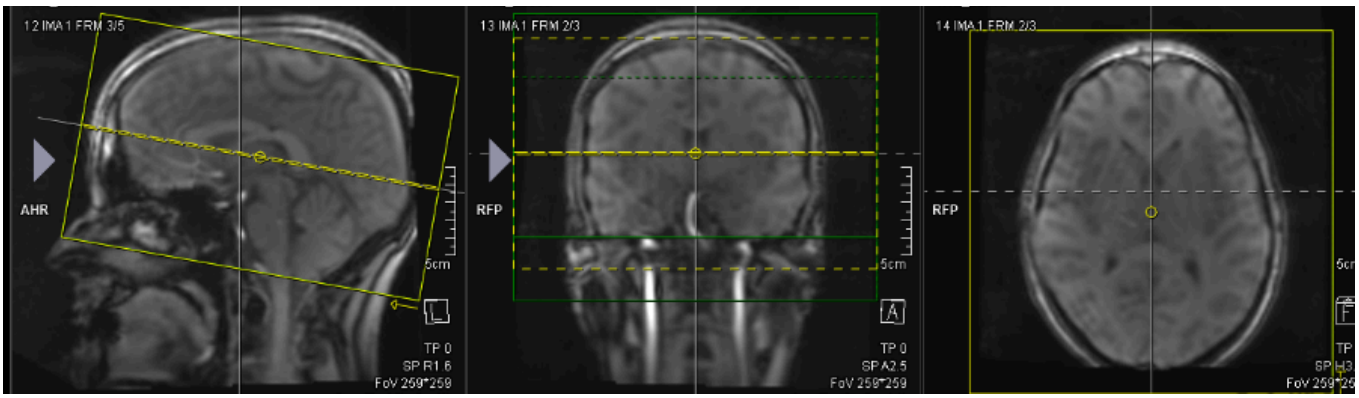
8. **gre_FA0_up** [corpus callosum, via COPYREFERENCE]
   - ☐ copy slice position from previous sequence
   - ☐ doublecheck RefVoltage = 0V
   - ☐ confirm RefVoltage when asked

9. **mp2rage_0p7iso_patientSpecific** [strict SAG & in ISO, NO PLANNING]
   - ☐ if SAR, switch to first level

10. **mp2rage_0p7iso_TrueForm** [strict SAG & in ISO, NO PLANNING]

11. **Manual adjustments for diffusion**
    - FWHM before shimming: _____ Hz
    - Run shim (2x GRE Brain with measure, calculate, apply)
    - FWHM after shimming: _____ Hz

12. **ep2d_diff_1p3iso_b2k5_PA** [ACPC via AutoAlign; screenshot only for DOUBLECHECKING]
    - ☐ save screenshot

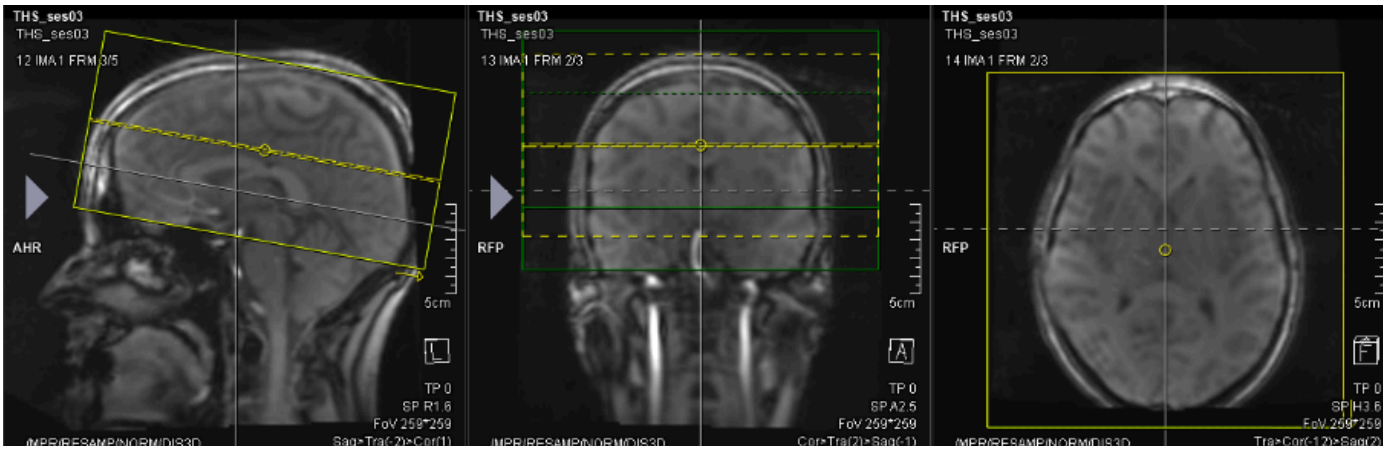


13. **ep2d_diff_1p3iso_b1k0_PA** [ACPC via COPYREFERENCE]
    - ☐ copy adjustment volume from ep2d_diff_1p3iso_b2k5_PA

14. **ep2d_diff_1p3iso_AP** [ACPC via COPYREFERENCE]
    - ☐ copy slice position from previous sequence
    - → DOUBLE CHECK phase encoding is "AP"
    - ☐ copy adjustment volume from ep2d_diff_1p3iso_b2k5_PA

15. **Manual adjustments for ep2d**
    - FWHM before shimming: _____ Hz
    - Run shim (2x GRE Brain with measure, calculate, apply)
    - FWHM after shimming: _____ Hz

16. **ep2d_bold_1p4mm with TASK** [ACPC via AutoAlign; might need shift up; screenshot only for DOUBLECHECKING]
    - ☐ save screenshot of FOV planning

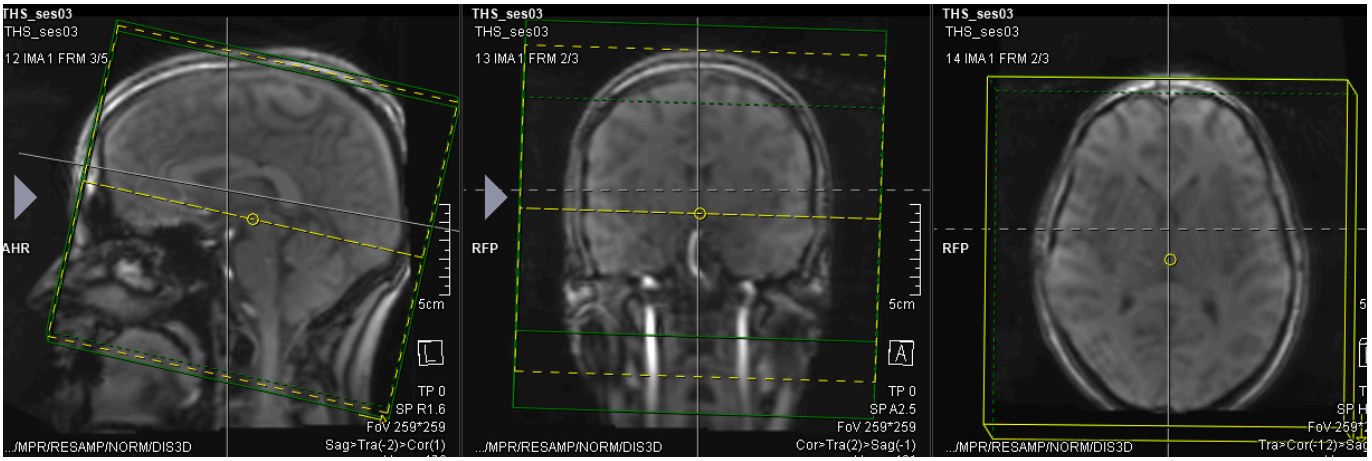


    - ☐ FYI: in Stanford the FOV had to be shifted headwards to cover the entire cerebrum
    - ☐ start THS_ep2d_TappingWithTrTiming_Movie_d3.py (see appendix for info)

17. **Manual adjustments for ep3d_vasoNIH**
    - FWHM before shimming: _____ Hz
    - Run shim (2x GRE Brain with measure, calculate, apply)
    - FWHM after shimming: _____ Hz

→ see next page

18. **dzne_ep3d_vasoNIH with Task, run 1** [ACPC whole brain, A>>P, might need shift down; screenshot only for DOUBLECHECKING]
    - ☐ save screenshot of FOV planning

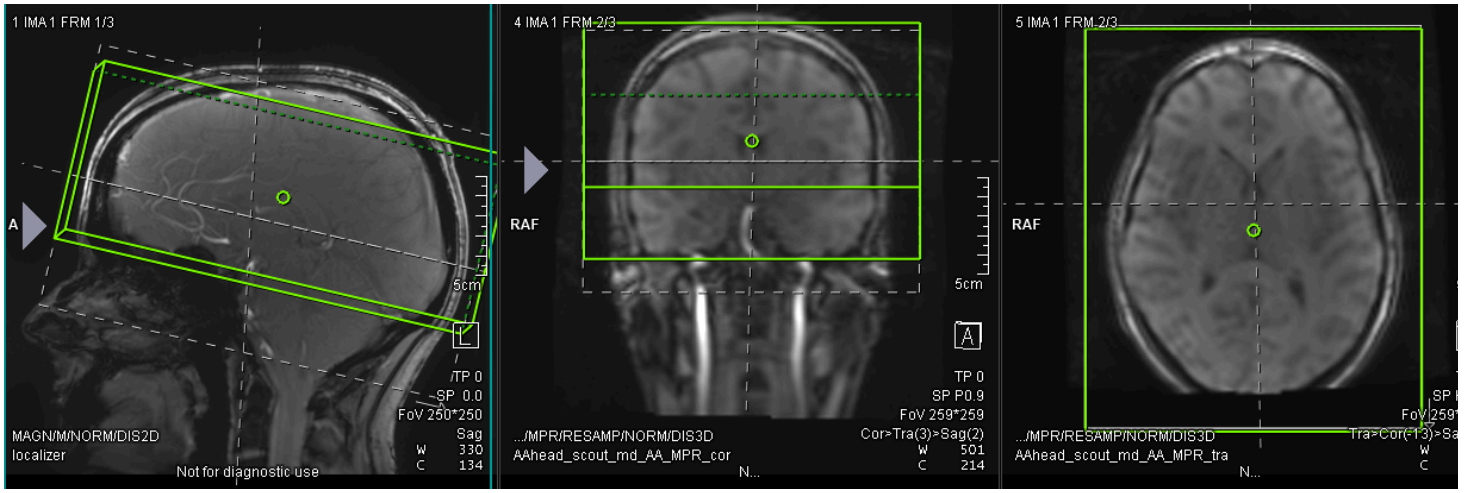


    - ☐ start THS_ep3d_TappingWithTrTiming_Movie_d3.py

19. **dzne_ep3d_vasoNIH with Task, run 2** [ACPC, whole brain via COPYREFERENCE]
    - ☐ start THS_ep3d_TappingWithTrTiming_Movie_d3.py

20. **Manual adjustments for meso-vein Frankfurt**
    - FIRST PLAN FOV & ADJUSTMENT VOLUME (see notes below)
    - FWHM before shimming: _____ Hz
    - Run shim (2x GRE Brain with measure, calculate, apply)
    - FWHM after shimming: _____ Hz

21. **dzne_ep3d_Frankfurt_6echoes** [ACPC, A>>P, **smaller Adj.Vol,** MANUAL PLANNING]
    - ☐ save screenshot of FOV planning

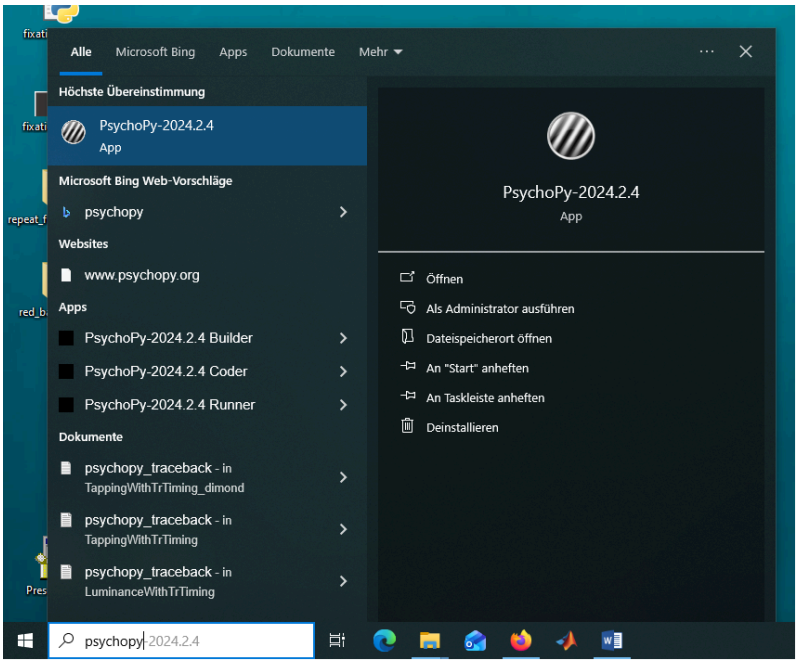


    - ☐ ACPC auto align, typically no adjustment of FOV required
    - ☐ IMPORTANT: Adjustment volume is smaller than the FOV but has the same angulation.
    - ☐ Adjustment volume WITHOUT EYES and covers most of the cerebrum
    - ☐ FOV covers entire cerebrum and parts of cerebellum
    - ☐ FYI: recon tasks ~15 min

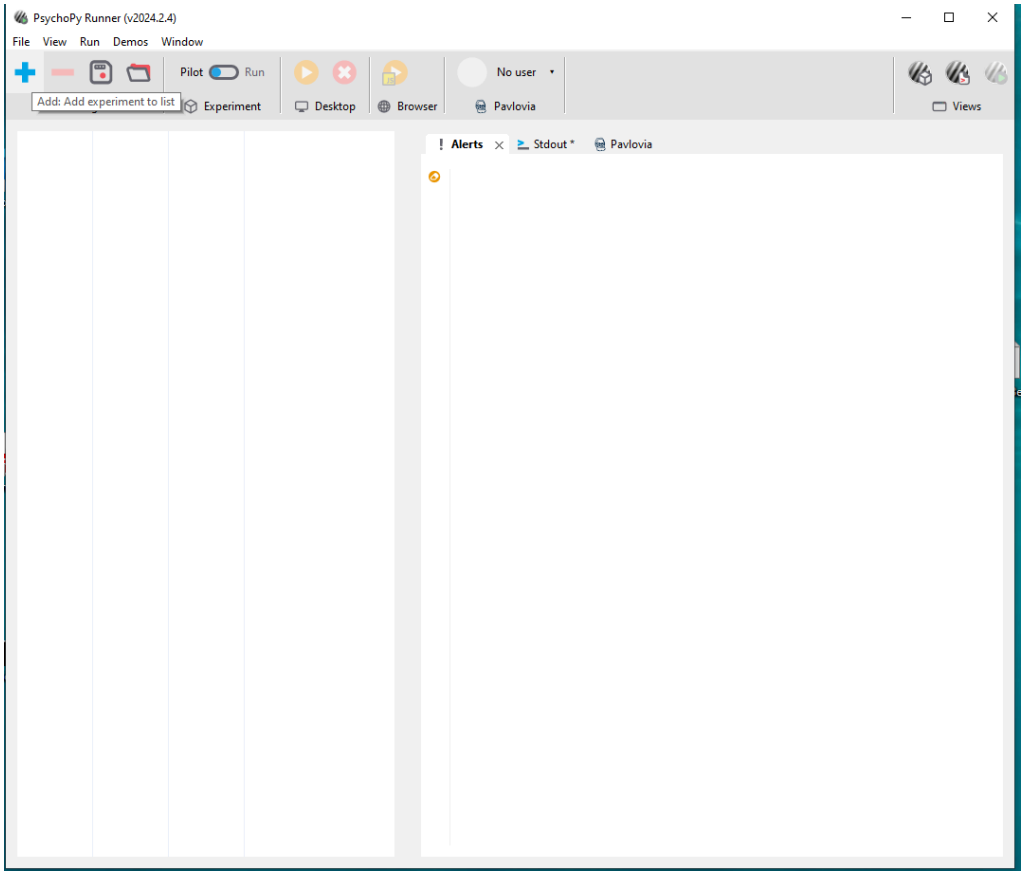
Appendix: How to start the paradigm in PsychoPy?

FYI: original code for THS here https://github.com/layerfMRI/Phychopy_git

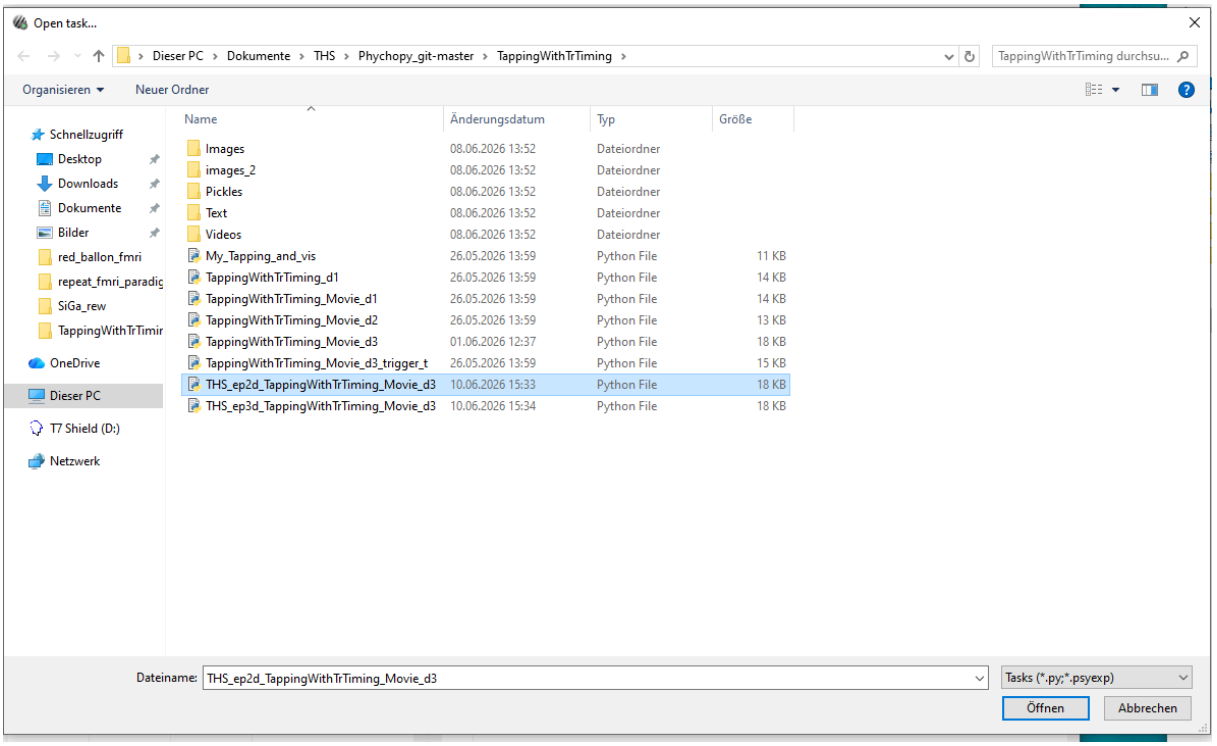
1.) Open PsychoPy



2.) Click “+” to add an experiment



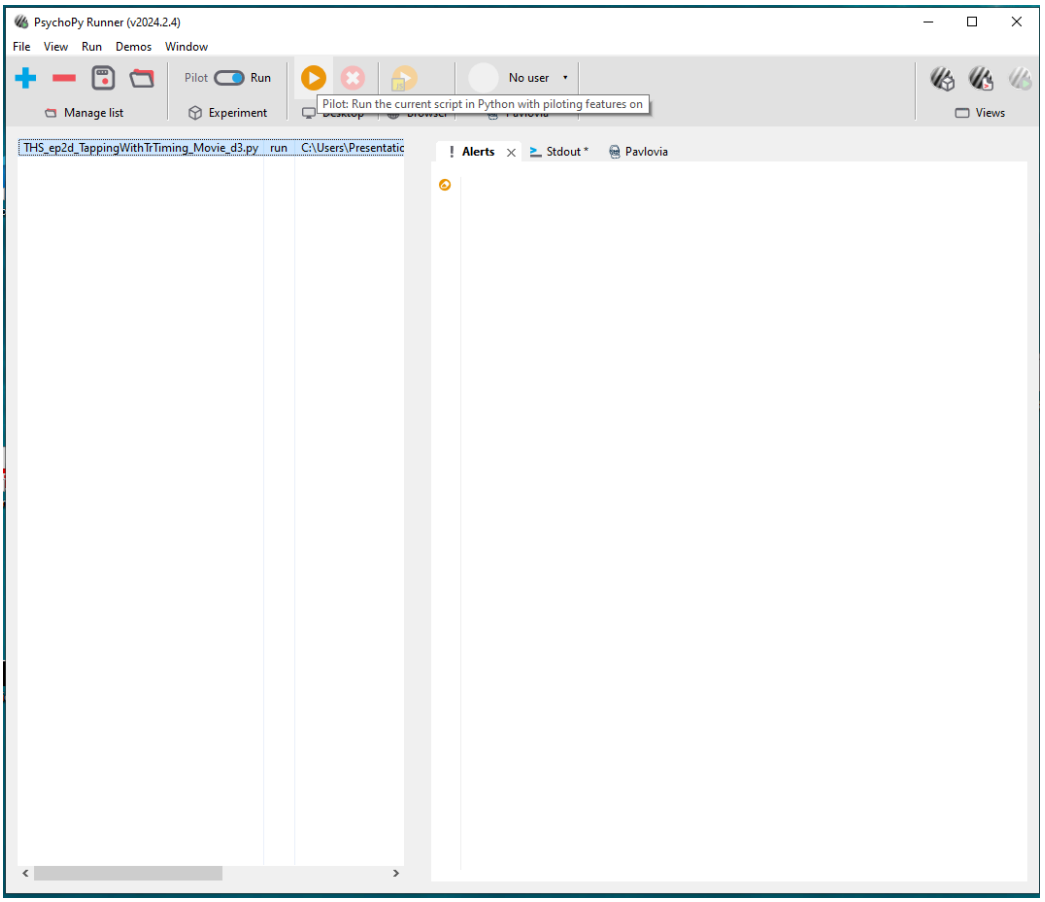
3.) Select file for task



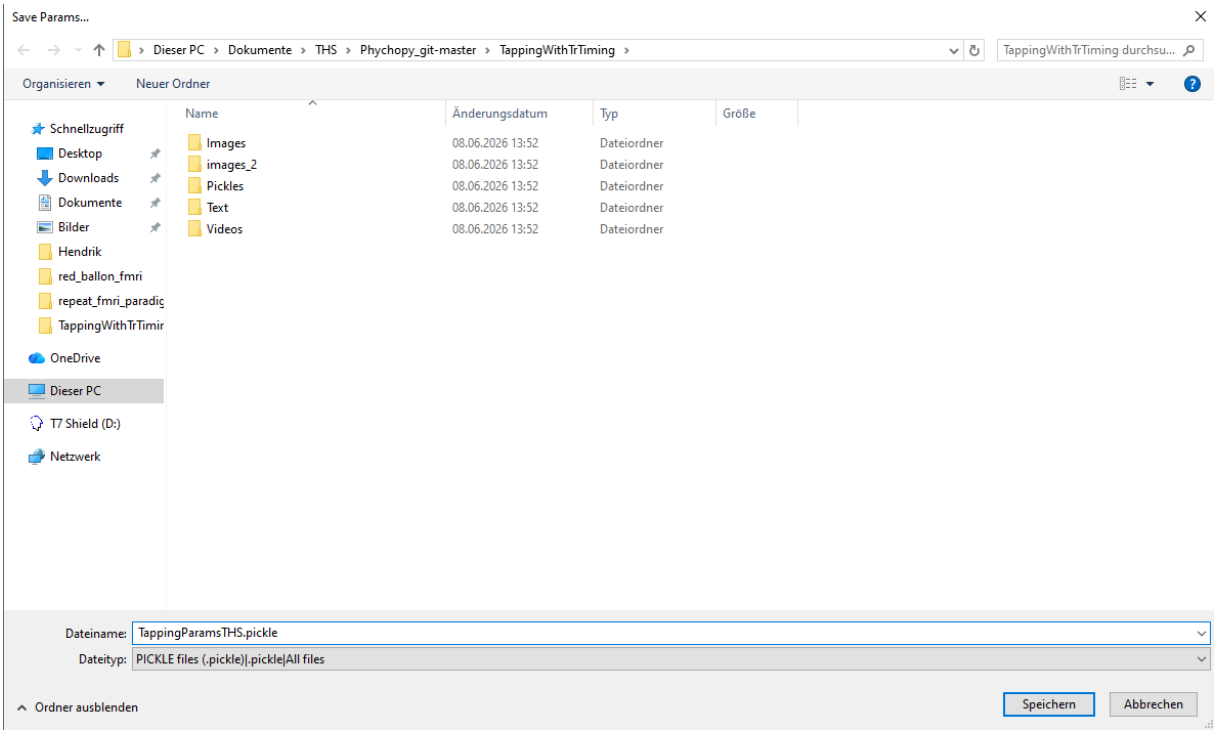
for 2D 20 TRs

for 3D 10 TRs

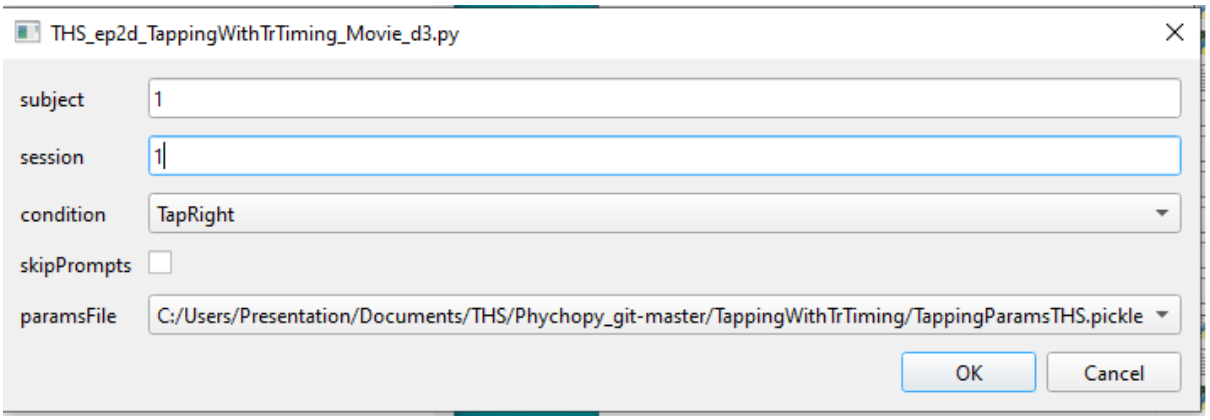
4.) Run the task by clicking the orange “play” button in the main PsychoPy window
IMPORTANT: the file you want to run needs to be selected/highlighted in blue



5.) Name the Log-File (PICKLE format) in the format “THS_<scan_date>_ep2d_run<Nr.>”, or “THS_<scan_date>_ep3d_run<Nr.>”, e.g. THS_2026_06_10_ep3d_run2 for a 3D EPI on the 10.06.2026, second run



6.) Enter subject ID=1, and session number=1 as each EPI will have an independent Log-File/PICKLE



7.) Press “Esc” or “q” twice to end the measurement if required